Dissertation Research Proposal

MSc Data Analytics

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1 Introduction

1.1 Background

The problem of sequential decision-making under uncertainty is at the heart of artificial intelligence, financial economics, and operations research. In practice, agents often encounter mechanisms whose internal dynamics are partially observable or fully hidden. These agents must make online decisions while simultaneously learning about the system.

For example, we have an agent that can use a machine to exchange two different types of candy, and the internal exchange rate of this machine is unknown. The agent deposits a quantity of candy A at every interaction and receives some quantity of candy B in return. But the true exchange rate is latent, and the observed output is noisy. The agent must therefore learn the latent rate through multiple interactions, deciding how much to exchange at each interaction.

1.2 Motivation

Learning under Uncertainty

The key challenge is that learning comes with a cost: information-gathering actions may degrade short-term performance, while greedy decisions might depend on erroneous beliefs.

The agent needs to experiment with deposit sizes to learn the exchange rate. A big candy A stash away will expose the true conversion rate quickly but runs the risk of an unfavourable payoff if the rate is low. Small deposits retain your holdings but will slow learning. The question is how much uncertainty the agent will accept for long-term growth.

Market Motivation

This dissertation was inspired by discussions about adaptive marketplaces, in which agents repeatedly invest resources under uncertain exchange rates. In such contexts, the actual exchange rate is seldom clear. Rather than observing the true rate directly, traders must learn under noisy feedback conveyed by repeated transactions. This leads to the principal question: in an uncertain, stochastic environment, how should a decision-maker learn the action-outcome mapping and allocate resources?

What you see here is a simplified market mechanism represented by the candy machine. Just as the true equilibrium exchange rate is not directly observable by traders, neither is the internal rate of the candy machine visible to the agent. It works by inferring the effective rate based on noisy transaction outcomes, which captures how uncertain real markets are while remaining conceptually tractable.

Broader Applicability

Although financial markets provide a concrete motivation, the essence of this problem is broader. Similar learning challenges arise in AMMs, resource allocation markets, distributed systems and electronic markets. In each case, agents make repeated decisions based on noisy outcomes generated by a partially unknown system.

Additive vs Multiplicative Objectives

The majority of classical reinforcement learning and bandit theory focuses on cumulative reward maximisation. However, in many economic and portfolio-growth settings, growth is multiplicative rather than additive: wealth compounds over time. In such scenarios, the geometric growth rate is a more appropriate notion of performance. The set of policies that maximise additive rewards is not necessarily the set that maximises long-term growth.

For example, in the candy case, if the agent has an incentive to maximise the total number of exchanges of candies, it may convert most of its holdings into one type because the per-exchange rate is high in some cases. This only leads to an imbalance in the portfolio: One type ends up dominating while another becomes rare. General value depends on both, which means that while this strategy could work if using an additive measure of utility, it will surely fail when a multiplicative one is applied. Balanced accumulation is designed for sustainable growth.

Proposed Exchange Model

We consider a simplified two-resource model in which an agent holds two commodities and interacts with a black-box exchange mechanism. The internal exchange rate is unobservable and subject to stochastic noise, and the agent must infer this parameter from repeated interactions. Unlike classical bandit problems that maximise additive reward, the purpose here is multiplicative: agents aim to maximise the product of their holdings, which captures balanced long-run growth that cannot be inferred from instantaneous wealth alone. By modelling the exchange result as a linear function of action (plus noise), the straightforward approach allows us to carry a objective. It gives a systematic way of studying exploration-exploitation trade-offs under compound growth criteria.

1.3 Problem Statement

We formalise the exchange example as a sequential decision process. At each time step t, the agent holds a portfolio (A_t, B_t) of two commodities. The agent chooses an action d_t representing the amount of commodity A deposited into the exchange mechanism. The mechanism returns a stochastic output

$$O_t = \ell^* d_t + \epsilon_t,$$

where ℓ^* is an unknown exchange parameter and $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$ are i.i.d. noise terms. The portfolio evolves according to

$$A_{t+1} = A_t - d_t, \qquad B_{t+1} = B_t + O_t,$$

subject to feasibility constraints $0 \leq d_t \leq A_t$. The agent observes only (d_t, O_t) and must learn ℓ^* through interaction while choosing actions that influence future portfolio states.

1.4 Research Questions

The central research question of this dissertation is:

More formally, we ask under what conditions sublinear estimation error implies sublinear growth regret under multiplicative portfolio dynamics.

More precisely, we investigate:

- **Question 1:** What is the sample complexity required to estimate the unknown exchange parameter ℓ^* with sufficient accuracy to guarantee near-optimal growth?
- **Question 2:** How does estimation error influence the long-term behaviour of the Nash welfare objective under compounding portfolio dynamics?
- **Question 3:** Can exploration strategies be designed that explicitly account for growth preservation rather than purely additive reward optimisation?

1.5 Objectives

To address the above question, this dissertation aims to:

- Formally characterise the statistical estimation problem of the unknown exchange parameter ℓ^* and derive bounds on the sample complexity needed to achieve ϵ-accurate estimation under stochastic feedback.
- Derive a theoretical link between parameter estimation error and the long-run growth of Nash welfare $U(A_T, B_T) = A_T B_T$, which accounts for the propagation of uncertainty through nonlinear compounded portfolio dynamics.
- Provide systematic exploration–exploitation design (e.g., confidence-bound-based policies) and analysis that includes explicit growth preservation, as opposed to pure additive reward optimisation.
- Conduct simulation experiments with varying noise levels and time horizons to empirically validate the proposed framework and evaluate robustness as well as practical feasibility.

2 Related Work

This work is situated at the intersection of stochastic bandit theory, growth-optimal portfolio dynamics and economic exchange mechanisms.

2.1 Stochastic Linear Bandits and Estimation

The UCB principle [1] and finite-time regret bounds for linear bandits [2, 3], among others, are the basis for learning under uncertainty. Standard models of regret are additive in nature, but more recent extensions have started looking at Thompson Sampling for linear payoffs [4] and other perspectives on convergence in a Bayesian context to parameters.

Gap: Existing theory mostly treats rewards as independent of the resource constraints present in the current state, and thus fails to consider how estimation error alters which future actions are feasible.

2.2 Growth-Optimal Theory and Multiplicative Objectives

Multiplicative growth stems from the Kelly Criterion [5], which defines the optimal growth strategy (the one that accumulates wealth over time) in a long game. Thorp [6] extended this to stock markets, connecting information rate to growth potential. In decision-making paradigms the Nash Welfare objective is a natural scalarization of multi-objective outcomes and came about from early axiomatic studies of choice [7].

Gap: Many economic models assume known parameters (ℓ^*). Compared to known-parameter growth optimization, stochastic learning-based growth transition is less explored.

2.3 Learning in Automated Market Mechanisms

The first is the notion of a "candy machine" exchange, which corresponds to modern Constant Function Market Makers (CFMMs); these are market operators where prices depend only on internal reserves [8]. In contrast to traditional models of asymmetric information [9], our concern is with statistical complexity of learning these exchange parameters from noisy, repeated interactions. Multi-objective reinforcement learning (MORL) provides a way to incorporate vector-valued outcomes (A_T, B_T) into sequential decision frameworks [10], but usually does not admit the compounding dynamics present in asset exchange.

Current CFMM literature (cite Angel etal 2023) has focused on liquidity, arbitrage and attaining kink or curve shaped volume or payout function while MORL centers around constructing Pareto frontiers but with limited success therefore not addressing the generative feature of how multiplicative growth can be maintained under parameter uncertainty.

2.4 Advanced Statistical Learning and Noise

Both standard RL and bandit theories assume sub-Gaussian noise that can be added; e. g. [11]. But multiplicative structures may create heteroscedasticity. Recent work in the setting of linear bandits with multiplicative noise [12] or online learning with multiplicative weights [13], provides tools for understanding how errors propagate through compounding systems.

Gap: Integrative analyses which connect these advanced statistical estimators to the long-term Nash Welfare of a resource-consecutive agent are missing.

3 Research Methodology

3.1 Model Overview

A Portfolio Management Setting. We model a sequential decision-making environment where an agent maintains a portfolio of two commodities. At each time step t, the agent has a state

$$(A_t, B_t)$$

where the variables A B refer to quantities of funds.

The agent selects an action

$$d_t \in [0, A_t]$$

where y_A indicates the amount of commodity A injected into a stochastic trading platform. We model the exchange mechanism to be linear function with unknown parameter:

$$O_t = \ell^* d_t + \varepsilon_t \tag{1}$$

where ℓ^* is an unknown fixed exchange parameter and $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$ indicates stochastic noise. This specification is in line with the stochastic linear bandit framework [2, 3].

3.2 Sequential Interaction Structure

In Figure 5, we present the closed-loop interaction of the agent and the exchange mechanism. At each round:

1. The agent sees its current portfolio state (A_t, B_t)
2. It picks a deposit d_t according to its belief about ℓ^*
3. The black-box mechanism outputs a noisy value O_t
4. The portfolio is adjusted as follows:

$$A_{t+1} = A_t - d_t, \quad B_{t+1} = B_t + O_t$$

This structure distinguishes it from classical additive bandit settings, as actions impact not only immediate feedback but also the future resource feasibility. This means exploration decisions impact the future action space directly.

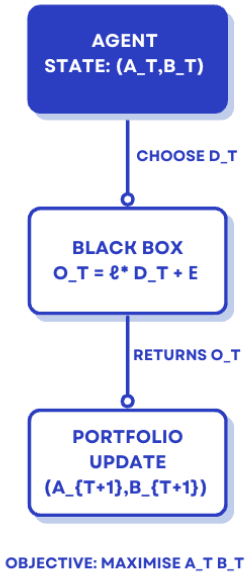

Figure 1: Agent-stochastic black-box exchange mechanism interaction in a sequential way. The agent observes noisy output and updates holdings based on current portfolio state and a selection of deposits.

3.3 Objective Function

The goal is to maximise terminal Nash welfare over a finite horizon T:

$$U(A_T, B_T) = A_T B_T.$$

This multiplicative objective can be seen as a scalarisation of a multi-objective portfolio outcome. In contrast to the additive reward formulations of classical reinforcement learning, performance is measured by multiplicative resource growth. Thus, sublinear additive regret does not imply near-optimal growth in general.

3.4 Experimental Design

The study is theoretical and simulation-based. No external empirical data is needed. Instead, they generate synthetic exchange environments according to the stochastic linear model.

Synthetic environments enable complete control over:

- Noise variance σ^2,
- Time horizon T,
- Exchange parameter ℓ^*,
- Initial portfolio (A_0, B_0).

This controlled environment allows for systematic comparison of learning behavior among different uncertainty regimes and objective specifications.

3.5 Proposed Methods

Parameter Estimation

The unknown exchange parameter ℓ^* is estimated using ordinary least squares:

$$\hat{\ell}_t = \arg\min_{\ell} \sum_{s=1}^{t} (O_s - \ell d_s)^2. \tag{2}$$

Under ordinary assumptions, this gives the closed-form estimator

$$\hat{\ell}_t = \frac{\sum_{s=1}^{t} d_s O_s}{\sum_{s=1}^{t} d_s^2}. \tag{3}$$

The statistical uncertainty is quantified using confidence bounds obtained from concentration inequalities. This bounds gives a confidence region for ℓ^* at each time step. The above estimation procedure is similar to the stochastic linear bandit framework and it follows [2, 3].

Exploration–Exploitation Strategy

We follow the principle of confidence-bound exploration inspired by UCB style algorithms [1]. At each timestep, the agent generates an optimistic estimate of the exchange rate that it can find in its confidence interval.

Whereas classical bandit formulations maximise expected reward in an additive sense, action selection in our framework involves both:

- estimation uncertainty about ℓ^*,
- current portfolio state (A_t, B_t).

The decision rule chooses deposits d_t to maximise an optimistic estimate of the growth in welfare:

$$U(A_{t+1}, B_{t+1}) = (A_t - d_t)\left(B_t + \tilde{\ell}_t d_t\right), \tag{4}$$

where $\tilde{\ell}_t$ is an upper confidence estimate of the exchange rate?

This results in a welfare-aware exploration policy: uncertainty is to be explored only as much as it positively contributes to expected multiplicative growth.

Deterministic Benchmark

To benchmark performance, we analyse the deterministic case where it ℓ^* is known and noise-free. To find the optimal deposit d^*, we need to solve:

$$\max_{d \in [0, A_t]} (A_t - d)(B_t + \ell^* d). \tag{5}$$

Taking the derivative with respect to d and setting it to zero yields a closed-form optimal deposit, subject to feasibility constraints.

3.6 Evaluation Strategy

Evaluation Metrics: Performance will be evaluated against the following metrics.

- **Estimation error:** $|\hat{\ell}_t - \ell^*|$
- **Sample complexity:** the minimal T such that

$$|\hat{\ell}_T - \ell^*| \leq \epsilon.$$

- **Growth regret:**

$$R_T^{\text{growth}} = U(A_T^*, B_T^*) - U(A_T, B_T),$$

  where (A_T^*, B_T^*) denotes the deterministic optimal portfolio obtained when ℓ^* is known.

- **Log-growth rate:** $\log(A_T B_T)$ to analyse compounding behaviour.

We will conduct simulation experiments to examine model convergence and robustness for different time horizons T, noise levels σ^2 and initial conditions.

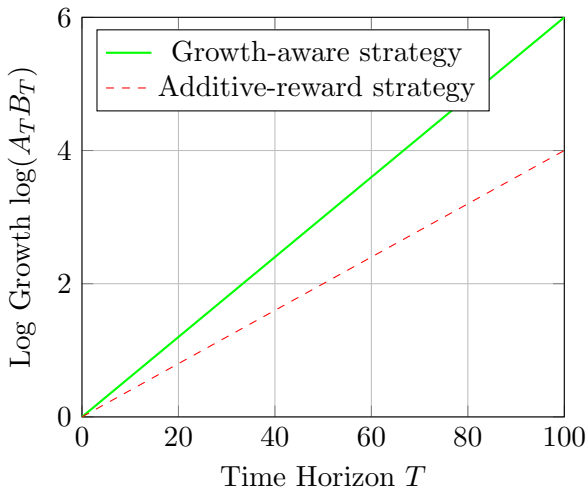

Figure 2: Theoretical illustration of asymptotic log-growth rates under different learning strategies. Empirical slopes will be estimated via simulation.

Slopes correspond to asymptotic log-growth rates for various learning strategies. Because log-growth linearises multiplicative welfare, steeper slopes correspond to better long-run exponential growth. Slopes will be estimated through simulation; the lines depict theoretical rates of growth.

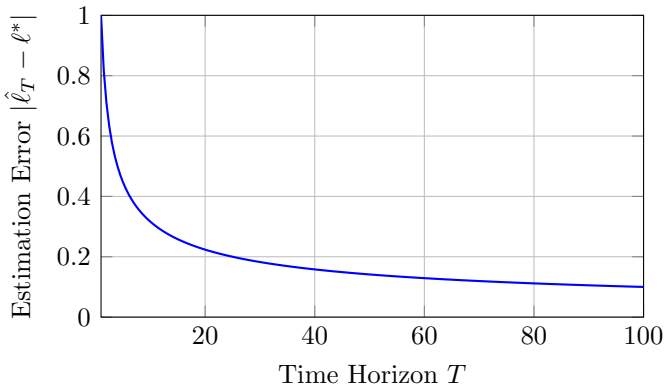

Figure 3: Illustrative convergence of estimation error as the number of interactions increases.

The convergence rate follows the order $O(1/\sqrt{T})$ with high probability under standard sub-Gaussian noise assumptions, consistent with classical stochastic linear bandit theory.

3.7 Feasibility

The proposed method is computationally inexpensive and can be implemented in Python using general numerical packages (NumPy, SciPy). The theoretical analysis builds upon established results in stochastic linear bandits and least-squares estimation.

The scope is still feasible for a master's dissertation timeline, as the model has one unknown parameter with controlled simulation environments. In terms of asymptotic proofs, our theoretical analysis will be focused on finding interpretable bounds instead of exhaustive approaches.

We further investigate whether sublinear estimation error translates into sublinear welfare regret under multiplicative growth dynamics.

4 Methodology and Design

The theoretical, algorithmic and experimental design of the proposed framework has been detailed in this section.

4.1 Theoretical Framework

We rigorously frame the stochastic exchange environment and multiplicative growth goal.

Now we consider an exchange mechanism modeled as a linear stochastic system with unknown parameter ℓ^*, and portfolio evolution dynamics given by (2) introduced in Section 3.

The aim is to optimize terminal portfolio product:

$$U(A_T, B_T) = A_T B_T.$$

We analyse:

- least-squares estimator statistical properties,
- confidence-bound constructions with respect to parameter uncertainty,

- how errors in estimation carry on, into long-run multiplicative growth.

This creates a formal connection between parameter learning and portfolio growth performance.

4.2 Algorithm Design

We develop exploration–exploitation strategies suited to multiplicative growth objectives.

At each time step, we compute confidence intervals for ℓ^* and choose deposits d_t to maximise optimistic growth:

$$\max_{d \in [0, A_t]} (A_t - d) \left(B_t + \tilde{\ell}_t d \right).$$

This gives a growth-aware version of classical confidence-bound approach.

We also implement comparison baselines:

- Deterministic optimal policy (ℓ^* known),
- Additive-reward UCB strategy.

4.3 Experimental Design

The simulation experiments take place in synthetic environments under controlled exchange parameter ℓ^*, noise variance σ^2, horizon T, and initial portfolio (A_0, B_0).

There are three dimensions along which performance is evaluated:

- Parameter estimation accuracy,
- Growth regret,
- Long-run log-growth rate.

The robustness and convergence behaviour is tested for multiple volatility regimes and portfolio configurations.

4.4 Analysis and Interpretation

Simulation results are jointly analysed with theoretical bounds to assess:

- whether sublinear estimation error guarantees sublinear growth regret,
- uncertainty admits exploration behaviour,

  The second property is the sensitivity to noise and initial portfolio imbalance.

This analysis illuminates the interaction of stochastic parameter learning and multiplicative portfolio growth.

4.5 Timeline

The project is planned to be completed by the end of July 2026. The detailed schedule is shown below.

Phase	Period	Key Tasks
1. Scope & Foundations	**01/11–30/11/2025**	Topic alignment, define main objective & constraints
2. Early Structure Draft	**01/12–20/12/2025**	Draft chapter structure, define core notation
3. Core Literature Review	**01/12/2025–15/01/2026**	Linear bandits, estimation bounds, growth objective references
4. Formal Model & RQs	**15/01–25/02/2026**	Formal model, assumptions, evaluation metrics
5. Theory I: Estimation Bounds	**01/03–25/03/2026**	Least-squares estimator, confidence bounds, sample complexity
6. Theory II: Link to Growth	**20/03–15/04/2026**	How estimation error affects terminal product/log-growth
7. Algorithm + Implementation	**01/04–10/05/2026**	Growth-aware exploration policy, coding simulation framework
8. Experiments + Writing + Finalisation	**01/05–31/07/2026**	Baseline + robustness experiments; results analysis; writing & revisions

Figure 4: Timetable

Figure 5: Gantt Chart

4.6 Risk Management

- **Risk 1: Theoretical Complexity.** In maths, it will likely not be easy to derive finite-time bounds connecting parameter estimation error to multiplicative welfare growth.

  Mitigation: We will start from simple settings (a linear model with a single parameter, Gaussian noise) and gradually generalise it to more complex situations. In the absence of tight bounds, we will favour interpretable upper bounds and asymptotic guarantees.

- **Risk 2: Algorithmic Instability with Compounding.** Multiplicative dynamics may amplify noise and lead to temporary portfolio imbalance during exploration.

  Mitigation: These extreme states of the portfolio will be mitigated by conservative exploration schedules and capped deposit policies. Under different regimes of noise, sensitivity analysis will ensure robustness.

- **Risk 3: Scope Expansion** Within the current writing timeframe, it may not be possible to extend this framework to a complete multi-objective Pareto form system.

  Mitigation: The main development work will continue to focus on the Nash welfare scalarisation framework, while Pareto extensions will be a future research direction, depending on time constraints.

5 Expected Impact

5.1 Academic Contribution

This dissertation contributes to the study of sequential learning under multiplicative growth objectives.

By associating linear parameter uncertainty with the dynamics of Nash welfare growth, this work generalises classical bandit analysis to non-additive regret. This framework provides a structured setting to analyse exploration–exploitation trade-offs under multiplicative growth objectives.

At a higher level of generality, the thesis presents a theoretical foundation linking bandit learning theory and growth-orientated economic objectives that can serve as a basis for future research activity on welfare-aware online decision-making.

5.2 Practical Relevance

On the practical side, the framework offers insights into making decisions in settings where returns compound over time and system parameters are plagued by uncertainty. These conditions emerge across adaptive marketplaces, automated exchange mechanisms, portfolio allocation and algorithmic trading systems.

Grasping the impact of statistical estimation error adjusted projections on long-run multiplicative outcomes may help tailor design parameters for more general learning agents, as well as allocate overall resources toward those systems that aim to maximise a typical output or utility trajectory instead. As a result, the findings may be applied to financial technology and automated market design.

5.3 Future Work

This work naturally leads to several extensions. First, the framework can be generalised to multi-parameter or multi-assets settings, where the exchange mechanism is based on higher-dimensional uncertainty. Second, the scalarised Nash welfare formulation could be substituted for a complete multi-objective Pareto analysis, enabling explicit characterisation of trade-off frontiers.

Finally, including adaptive or time-varying exchange rates will allow to study non-stationary environments, getting even closer to the actual market dynamics and bandit learning.

6 Conclusion

This proposal presents a study of sequential decision-making under parameter uncertainty in stochastic exchange environments. If classical bandit theory considers only additive cumulative reward, many real-world growth processes make more sense in a multiplicative way. This observation justifies the interest in welfare-based objectives under learning dynamics.

Using a linear stochastic system with unknown parameters to model the exchange mechanism preserves faithful analytical tractability while allowing for a nonlinear compound objective

through Nash welfare maximisation in the dissertation. We outline a framework for establishing formal links between statistical estimation guarantees and long-term growth performance.

Using theoretical analysis, accompanied by controlled simulation experiments, this project aims to elucidate the ways in which exploration–exploitation trade-offs manifest when performance is defined in terms of multiplicative growth rather than additive reward. The results aim to clarify how learning dynamics interact with multiplicative growth objectives in stochastic exchange environments.

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